Hydrogen
and
its Compound

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Hydrogen and Its Compound

KNOW ABOUT HYDROGEN

Atomic number of Hydrogen is one and its electronic configuration is $1s^1$.

It is the **1st element** of the periodic table as elements in the periodic table are organized according to their electronic configurations. But its position is still doubtful as its properties resemble with both alkali metals and halogen.

Hydrogen **resemblance to alkali metals.** It forms oxides, halides and sulphides like the alkali metals.

It has **high ionization enthalpy and does not show metallic** characteristics under normal conditions.

Hydrogen shows similarity with halogens as:

It forms large number of covalent compounds but its reactivity is very low as compared to halogens.

It is a diatomic molecule, combines with other elements to form hydrides

Its ionisation energy is similar with them. Ionisation energy of Lithium(Li) is 520 $kJmol^{-1}$, Fluorine(F) is 1680 $kJmol^{-1}$ and that of Hydrogen is 1312 $kJmol^{-1}$.

Note :

It is concluded that Hydrogen shows similar properties to both alkali metals & halogens and has some differences from them as well. That is why the relevant question is asked that where should it be placed in the periodic table?

Points to remember

- Hydrogen is the most abundant element in the universe.
- Electronic configuration: $1s^1$
- Group Numbered: 1
- Period Number: 1
- Hydrogen(H) is Non-metal, dimeric form (H_2)
- Hydrogen(H) properties are not matched with other elements.

Q1 **Explain why hydrogen is best placed separately in the periodic table of elements?**

Sol. Electronic configuration of hydrogen resembles alkali metals (group I) and halogen (VII) and it shows reactions and properties similar as well as dissimilar to the elements of these two groups.

Note :

At very high pressure, hydrogen has been found to behave as metal so its position with alkali metals is more justified than with halogens.

Different form of Hydrogen (H):

A. Isotopes
B. based on reactivity
C. Nuclear Isomers.

(A) Isotopes :

Hydrogen (H) has three isotopes :
Protium $^{1}_{1}H$**, deuterium** $^{2}_{1}H$ **or D and tritium** $^{3}_{1}H$ **or T.** These isotopes have different number of neutrons in the nucleus and they show different physical properties.

	H_2 (Protium)	D_2(Deuterium)	T_2(Tritium)
Abundance	H_2 >	D_2 >	T_2
Proton	1	1	1
Neutron	0	1	2
Mass Number	1	2	3
Molecular weight	H_2 <	D_2 <	T_2
Melting point	H_2 <	D_2 <	T_2
Boiling point	H_2 <	D_2 <	T_2
Enthalpy of vaporisation	H_2 <	D_2 <	T_2
Bond length	H_2 >	D_2 >	T_2

(B) Based on Reactivity

(i) Simple atomic Hydrogen: It is formed by simple dissociation of Hydrogen.

$$H_2 \xrightleftharpoons{3000^{\circ}C - 5000^{\circ}C} 2H$$

$$H_2 \xrightarrow{\Delta} 2H \text{ (Atomic Hydrogen(H))}$$

Atomic Hydrogen(H) energy is more and stability is less and reactivity also more.

(ii) Nascent Hydrogen: Hydrogen forms at the instant is known as nascent Hydrogen means Hydrogen at the moment of its birth it known as nascent Hydrogen.
It is produced only by some specific chemical reaction.

Points to remember

According to (NCERT) $H_2 = D_2$ Bond length

The Reason for average distance H_2 is more because T_2 is heavy weight so Vibration in T_2 is less And H_2 is Lightweight so Vibration in H_2 is more.

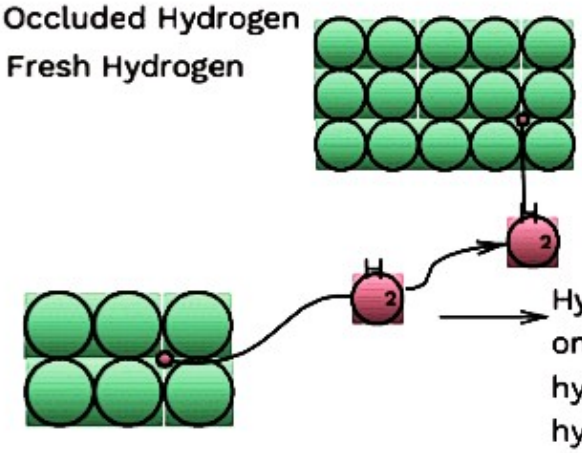
Bond lenght $H_2 > D_2 > T_2$

Bond energy $H_2 < D_2 < T_2$

Bond energy (m ore in T_2) because T_2 is heavier weight vibration occur less in T_2.

(iii) Occluded Hydrogen (fresh Hydrogen) :

Occluded Hydrogen
Fresh Hydrogen



Transition Metal
d-Block element
reactivity is low

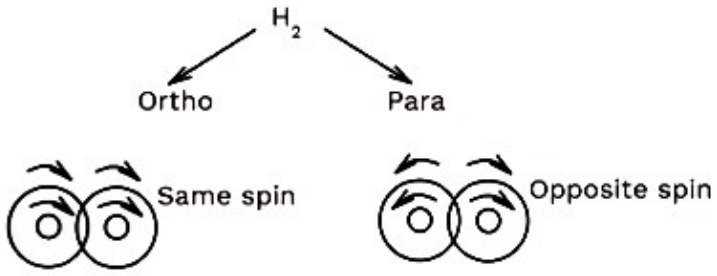
Hydrogen molecule is adsorbed on the metal surface, then this hydrogen is known as occluded hydrogen.

Interstitital site = voids
Interstitial Hydride

Order of reactivity: Atomic Hydrogen > Nascent Hydrogen > Occluded Hydrogen

(C) Nuclear Isomers :

(H_2) have two nuclear isomer



(a) Ortho hydrogen	It is the diatomic form of Hydrogen having same spin of nucleus.
(b) Para hydrogen	It is the diatomic form of Hydrogen having opposite spin of nucleus.

As Ortho Hydrogen has same spin of nucleus, they will repel each other and due to repulsion its internal energy increases. So ortho Hydrogen has more internal energy.

Stability of ortho & para hydrogen

Stability of ortho and para hydrogen depends **upon temperature** condition.

Preparation of Dihydrogen:

(1) Laboratory preparation of dihydrogen H_2

Reaction of acid with metal

$$\underset{\text{Active metal}}{Zn} + \underset{\text{Non-oxidising agent}}{\text{dil.HCl}} \longrightarrow ZnCl_2 + H_2\uparrow$$

Note :

- Cu/Ag + Acid $\longrightarrow$ (No reaction) = Then acid is non oxidising acid

  $Zn + \text{dil. } H_2SO_4 \longrightarrow ZnSO_4 + H_2\uparrow$

  $Cu + \text{dil. } H_2SO_4 \longrightarrow ZnSO_4 + SO_2 + H_2O$

- $Zn + \text{Conc. } H_2SO_4 \longrightarrow ZnSO_4 + SO_2 + H_2O$

  Zinc react conc. H_2SO_4 to give $ZnSO_4$. It is called oxidizing acid

  $Cu + \text{conc. } H_2SO_4 \longrightarrow CuSO_4 + SO_2 + H_2O$

In metal Reactivity Series, A metal below H_2 react with acid.

M + Acid $\longrightarrow$ reaction, then acid is oxiding acid.

M + acid $\longrightarrow$ no reaction, then acid is non oxidising acid

Imortant Point:

- In the reactivity series, metal below H_2 do not react with dilute acid. (Non-oxidizing acid).
- In the reactivity series, metal above H_2 react with dilute and concentration acid.
- **$KMnO_4$** is a good oxidising agent.

  Oxidising strength of $KMnO_4$:

  $H^+ > H_2O > OH^-$(Medium)

  $K_2Cr_2O_7$ → (Potassium dichromate)

  $K_2Cr_2O_7$ → Most commonly used oxidising agent only in acidic-medium.
- H_2O_2 is a good oxidising agent in an acidic and neutral but weak oxidising agent in a basic medium.

Points to remember

- At Room temperature 75% ortho and 25% para form is present.
- On decreasing the temperature, % of ortho H_2 decreases and % of para H_2 increases.
- Same spin of nuclei leads to more energy and opposite spin of nuclei leads to low energy and become more stable. Due to opposite spin it is exothermic.

Points to remember

Li		
K	Cr	
Ba	Fe	Cu
Sr	Cd	Hg
Ca	Co	Ag
Na	Ni	Au
Mg	Sn	Pt
Al	Pb	
Mn	(H_2)	
Zn		

$Ca + H^+ \longrightarrow Ca^{2+} + H_2\uparrow$

$Cu + H^+ \longrightarrow$ No reaction

Cu + dil HCl $\longrightarrow$ No reaction (non oxiding)

As, $H^+ \uparrow$ increases oxidising nature increases $\uparrow$.

Oxyacid (Oxidising and Non-oxidising)

(+3) (+4) (+5)

H_3BO_3 H_2CO_3 HNO_3

$\longrightarrow$

As we move from left to right

Electronegativity increases $\Rightarrow$

It means, oxidising nature also increases

(+3) H_3BO_3	(+4) H_2CO_3	(+5) HNO_3
Non-oxidising agent $B(OH)_4^-$ $X + 4(-1) = -1$ $X = +3$ Boron show +3 oxidation state So, they are a non-oxidising agent	Non oxidising agent $\overset{(+2)}{CO}/\ \overset{(+3)}{H_2C_2O_4}$ Reducing agent Carbon stable oxidation state (+4). So, they also show non-oxidising agents but sometimes reducing agents also.	• 70% Concentration HNO_3 (Oxidising agent) • 20% dilute HNO_3 (Oxidising agent) • 6% very dilute HNO_3 (Non-oxidising agent) • 2% very-very dilute HNO_3(Non-oxidising agent) • HNO_3 is increasing then H^+ also increases so, oxidising nature also increases. • $\overset{+2}{HNO_2}$ is oxidising and reducing agent both.

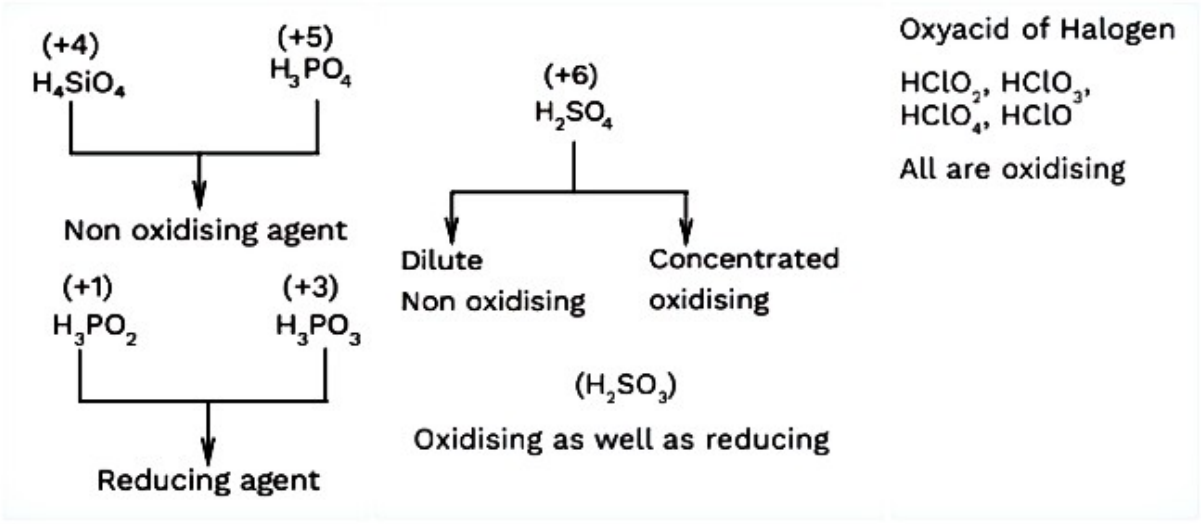
H_4SiO_4 H_3PO_4 H_2SO_4 Oxyacid of halogen

$\longrightarrow$

Left Right

Electronegativity increases (EN)

It means oxidising nature also increases.



Preparation of H_2 (Laboratory methods)

(1) Active Metal + Non-oxidising acid $\rightarrow H_2\uparrow$
It is mostly manufactured by the reaction of zinc (granulated) with dilute hydrochloric acid.
$Zn + \text{dil. } H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$

(2) Active Metal + Water $\rightarrow H_2\uparrow$
$Na + H_2O \longrightarrow NaOH + H_2\uparrow$

(3) It is also prepared by the reaction of zinc with aqueous alkali.
$Zn + NaOH \longrightarrow$

Amphoteric Metal + NaOH $\rightarrow$
Zn, Be, Al, Ga, Sn, Cr^{+3}, V^{+5}, Sb^{+3}, As^{+3}, Pb^{+3}

$$Zn + H_2O \longrightarrow Zn(OH)_2 + H_2\uparrow$$

$$\underset{\text{Lewis acid}}{Zn(OH)_2} + \underset{\text{Base}}{2NaOH} \longrightarrow \underset{(Na_2ZnO_2.2H_2O)}{Na_2Zn(OH)_4} + H_2\uparrow$$

$$Al + NaOH \longrightarrow \underset{\text{Sodium Aluminate}}{NaAlO_2} + H_2\uparrow$$

$$Al + H_2O \longrightarrow Al(OH)_3 + H_2\uparrow$$

$$\underset{\text{Acid}}{Al(OH)_3} + \underset{\text{Base}}{NaOH} \longrightarrow \underset{(NaAlO_2.2H_2O)}{NaAl(OH)_4} \text{ or } + H_2\uparrow$$

Points to remember

Oxidising Nature
Concentrate HNO_3 >
Concentrate H_2SO_4

$$Be + NaOH \longrightarrow Na_2BeO_2 + H_2 \uparrow$$

$$Be + H_2O \longrightarrow Be(OH)_2 + H_2 \uparrow$$

$$Be(OH)_2 + 2NaOH \longrightarrow \underset{\substack{(Na_2BeO_2.2H_2O)\\ \text{Sodium Berylate}}}{Na_2Be(OH)_2} + H_2 \uparrow$$

$$Sn + NaOH \longrightarrow Na_2SnO_2 + H_2 \uparrow$$

$$Sn + NaOH \longrightarrow Sn(OH)_2 + H_2 \uparrow$$

$$Sn(OH)_2 + NaOH \longrightarrow \underset{\text{Sodium Stannite}}{Na_2Sn(OH)_4}$$

$$Cr + NaOH \longrightarrow \underset{\text{Sodium Chromite}}{NaCrO_2} + H_2 \uparrow$$

$$Pb + NaOH \longrightarrow \underset{\text{Sodium Plumbite}}{Na_2PbO_2} + H_2 \uparrow$$

(4) Ionic Hydride + $H_2O \rightarrow$

Example

$$CaH_2 + H_2O \longrightarrow Ca(OH)_2 + H_2 \uparrow$$

$Ca^{+2}H^- \; H^+OH^-$

Q2 **Which pair of substances give same product?**

(A) CaO, CaH_2 **(B) Ca, CaH_2** **(C) Ca, CaO** **(D) None of these**

Sol. **(C)**

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

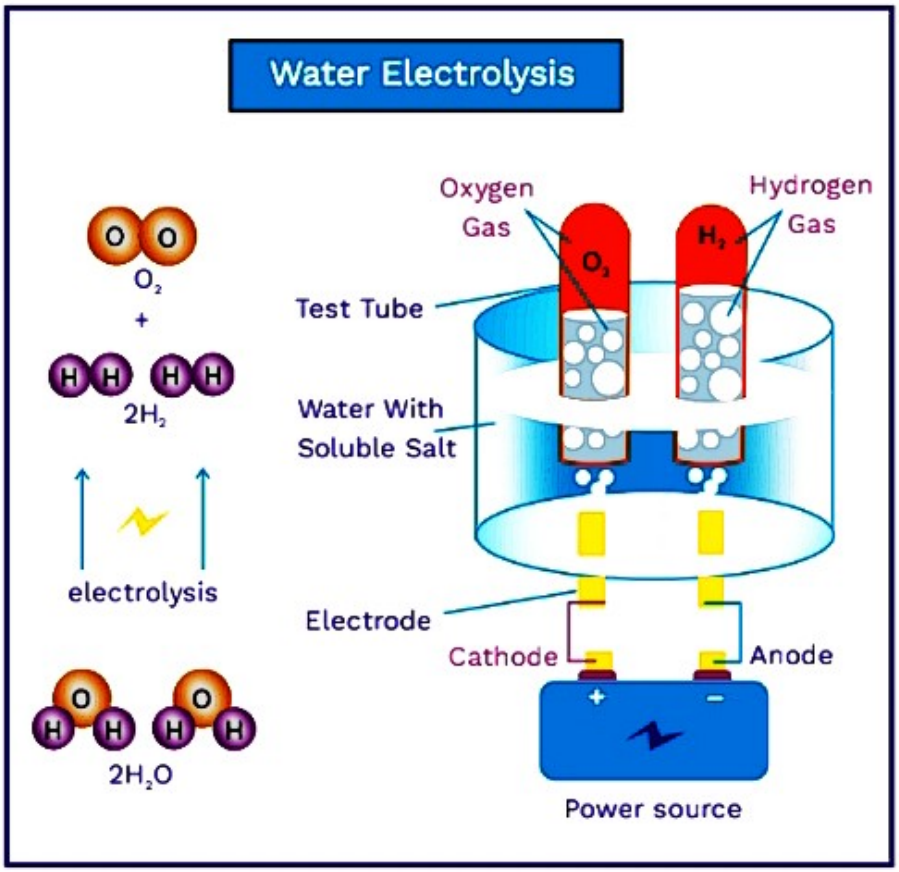
$$Ca + H_2O \longrightarrow Ca(OH)_2 + H_2\uparrow$$

Industrial Method:

The commonly utilized processes are given below:

(i) Hydrogen is obtained on the **Electrolysis of acidified water using platinum electrodes.**

$$2H_2O\ (\ell) \xrightarrow[\text{Trances of acid / base}]{\text{Electrolysis}} 2H_2(g) + O_2(g)$$



(ii) **Electrolysis of $Ba(OH)_2$ solution :**

High purity hydrogen is obtained by electrolysing the warm aqueous solution of barium hydroxide.

At Anode : $OH^- \rightarrow O_2 + 2e^-$

At cathode : $2H^+ + 2e^- \rightarrow H_2\uparrow$

(iii) **Electrolysis of Brine solution :**

It is obtained as a by-product in the manufacturing of sodium hydroxide and chlorine(Cl_2) by the electrolysis of brine solution.

The reactions that take place during the electrolysis are:

At anode : $2Cl^-(aq.) \longrightarrow Cl_2(g) + 2e^-$

at cathode: $2H^+ + 2e^- \longrightarrow H_2(g)$

The overall reaction is

$2Na^+(aq) + 2Cl^-(aq) + 2H_2O(\ell) \rightarrow Cl_2(g) + H_2(g) + 2Na^+(aq) + 2OH^-(aq)$

(iv) **Reaction of steam on hydrocarbons or coke at high temperatures in the presence of catalyst yields** Hydrogen(H_2).

$$C_nH_{2n+2} + nH_2O \xrightarrow[Ni]{1270\ K} nCO + (2n + 1)\ H_2$$

Example :

$CH_4(g) + H_2O(l) \xrightarrow[Ni]{1270\ K} CO(g) + 3H_2(g)$

The mixture of CO and H_2 is known as **water gas.** As this mixture of CO & H_2 is utilized for the synthesis of methanol, it is also known as synthesis gas or **'syngas'**.

Nowadays 'syngas' is obtained from saw-dust, sewage, scrap wood, newspapers etc. The process of generating 'syngas' from coal is called coal gasification.

$C(s) + H_2O(g) \xrightarrow{1270\ K} CO(g) + H_2(g)$

The production of dihydrogen can be increased by reacting syngas with steam in the presence of iron chromate as catalyst (Fe_2O_3 and Cr_2O_3).

$CO(g) + H_2O\ (g) \xrightarrow[catalyst]{673\ K} CO_2(g) + H_2(g)$

Carbon dioxide(CO_2) is seperated by scrubbing using sodium arsenite solution. This is called **water-gas shift reaction (Bosch Process)**.

Note:

In current scenario 77% of the industrial dihydrogen is generated from petro-chemicals, 18% from coal, 4% from electrolysis of aqueous solutions and 1% from other sources.

Physical and Chemical Properties of dihydrogen:

1. Physical Properties :

H_2 is colourless, Diatomic, Diamagnetic, Nonpolar, insoluble in water, Highly Inflammable $H_2 + O_2 \rightarrow H_2O$ (Explosive Reactions)

2. Chemical Properties :

(i) With S-block metal

$Na + H_2 \rightarrow NaH$

$Ca + H_2 \rightarrow CaH_2$(Hydrolith)

(ii) With Halogens

$X_2 + H_2 \rightarrow HX$

Bond Energy $\rightarrow HF > HCl > HBr > HI$

$\Delta H_f \rightarrow$ Change in enthalpy

$-\Delta H_f \rightarrow$ Release energy

$\Delta H_f : HF > HCl > HBr > HI$

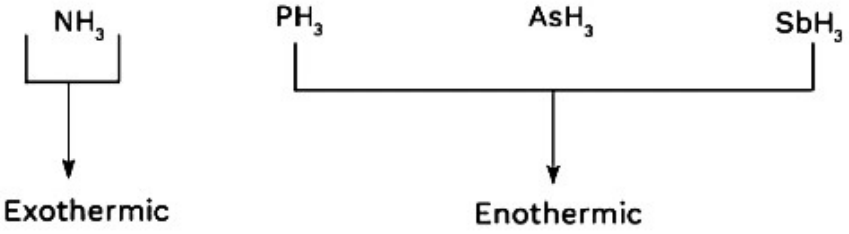
Reactivity $\rightarrow F_2 > Cl_2 > Br_2 > I_2$

Florine reacts with the hydrogen in the dark and chlorine, bromine, Iodine needs to be a catalyst.

(iii) With group 16 elements $E + H_2 \rightarrow H_2E$



(iv) With group 15 elements $E+H_2 \rightarrow EH_3$



(v) With low reactive metal ions.
Reactivity < H_2

$$Cu^{2+}_{(aq)} + H_2 \rightarrow Cu \downarrow + H^+$$

Q3 **How is di-hydrogen prepared:**
(A) from water by using a reducing agent?
(B) in the laboratory in pure form?
(C) from hydrocarbons?

Sol. (A) By reduction of water with Na or by electrolytic reduction.

$$2Na + 2H_2O \rightarrow 2NaOH + H_2$$

$$2H_2O \xrightarrow{\text{electrylysis}} 2H_2 + O_2$$

(B) By dissolution of NaH or CaH_2 in H_2O

$$NaH + H_2O \longrightarrow NaOH + H_2$$

$$CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + H_2$$

(C) By the action of-stream.

$$C_nH_{2n+2} + 2H_2O(g) \xrightarrow[\text{Catalyst}]{1270K} nCO + (2n+1)H_2$$

Q4 **What is understood by hydrogen economy?**

Sol. The production, storage and transportation of energy in the form of liquid or gaseous hydrogen as a fuel in view of the facts that fossil fuels are limited, electricity can't be stored and use of nuclear power is restricted.

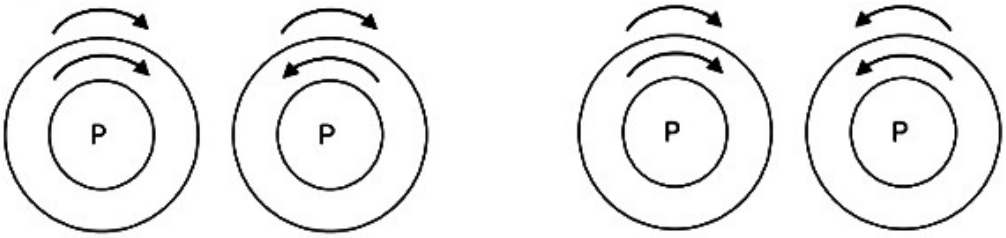
Q5 **Is it correct to say that hydrogen can behave as a metal? State the conditions under which such behaviour can be possible.**

Sol. Yes like metals it reacts with halogens, oxygen and other non-metallic elements.

$$H_2 + Cl_2 \longrightarrow 2HCl$$

Q6 **How many nuclear spin isomers of dihydrogen are know? Compare any two properties of these.**

Sol. Ortho and para hydrogen are allotropes. These are better called nuclear spin isomers of dihydrogen. Thermal conductivity of para H_2 is more, m.pt. of para H_2 is less.



Ortho H_2 Para H_2

Q7 **Hydrogen is a gas at ordinary temperature. How it can be stored as a solid. Explain.**

Sol. H_2 can be stored as a solid in the form of solid interstitial hydrides or in the recently discovered tube-shaped molecules called carbon nanotubes.

Q8 **What is the most important method for industrial preparation of H_2.**

Sol. The steam-hydrocarbon reform process

$$CH_{4(g)} + H_2O_{(g)} \xrightarrow[Ni]{1100°C} CO_{(g)} + 3H_{2(g)}$$

$$CO_{(g)} + H_2O_{(g)} \xrightarrow{400°C} CO_{2(g)} + H_{2(g)}$$

$$CO_{2(g)} + 2OH^-_{eq.} \longrightarrow CO^{2-}_{3\,eq.} + H_2O_{(g)}$$

Q9 **What are the advantages in using hydrogen as a fuel?**

Sol. It's availability in huge amount and lighter nature to store in liquid form.

Q10 **When fused calcium hydride is electrolysed, hydrogen is liberated at anode. Why?**

Sol. The H exist as H^- in CaH_2. During electrolysis, H^- ions are oxidized at anode to liberate H_2.

$$CaH_2 \longrightarrow Ca^{2+} + 2H^-$$

Anode : $2H^- \longrightarrow H_2 + 2e^-$ (Oxidation)

Cathode : $Ca^{2+} + 2e^- \longrightarrow Ca$ (Reduction)

Q11 **The electrolysis of water manufacturing hydrogen gas is always carried out in presence of acid (H_2SO_4) or alkali (KOH), yet no SO_4^{2-} or K^+ ions are discharged, why?**

Sol. The discharge potential of SO_4^{2-} > discharge potential of OH^- and the discharge potitial of K^+ > discharge potential of H^+.

Note :

Higher is discharge potential, lesser is tendency to get discharged.

12. **When zinc is treated with dil. Acetic acid. Explain why Hydrogen gas is evolved more quickly?**

Sol. HCl being strong electrolyte, it ionises completely to give higher concentration of H^+ and Cl^- ions. Acetic acid is feebly ionised to given less concentration of H^+ ions as it is weak electrolyte. These are the H^+ ions which on reduction give H_2.

Q13 **Explain, why hydrogen is discharged at anode when sodium hydride in fused state is electrolysed.**

Sol. Due to electrovalent nature of the sodium hydride, hydrogen is present as an anion H^-. On electrolysis, oxidation takes place at anode and hydrogen is discharge.

$$NaH \rightleftharpoons Na^+ + H^-$$

Cathode: $Na^+ + e \longrightarrow Na$

Anode: $H^- \longrightarrow H^+ + e^-$

$2H^+ \longrightarrow H_2 \uparrow$

Type of hydride

- **Ionic Hydride**
  - Saline (Salt like)
  - {S-block, Lanthnoid}
  - {Except Be & Mg}
- **Interstitial**
  - d-block Hydride
  - They are not simple compound non-Stoichiometric Compound
  - Formula is not In Simple ratio there Is no whole number
  - Like Fe_3C, FeO
- **Covalent**
  - e^- defficient $< 8e^-$
    - group 13
    - BeH_2, MgH_2
  - e^- precise $= 8e^-$
    - Absent lone pair
    - group 14
    - CH_4
  - e^- rich $= 8e^-$
    - Lone pair present
    - group 15, 16, 17,
    - NH_3, H_2O, HF

Water

It is the main constituent of all living organisms. Human body contains about 65% water and a few plants have as very much as 95% water. It is a solvent of great importance called universal solvent. There is diverse distribution of water over the earth's surface.

Points to remember

If you want more than 8 electrons then we used d-orbited and d-orbital is faraway so, d-orbital come closer, then we need a positive charge on central atom, so PF_5 exist but PH_5 does not exist.

Physical Properties of Water

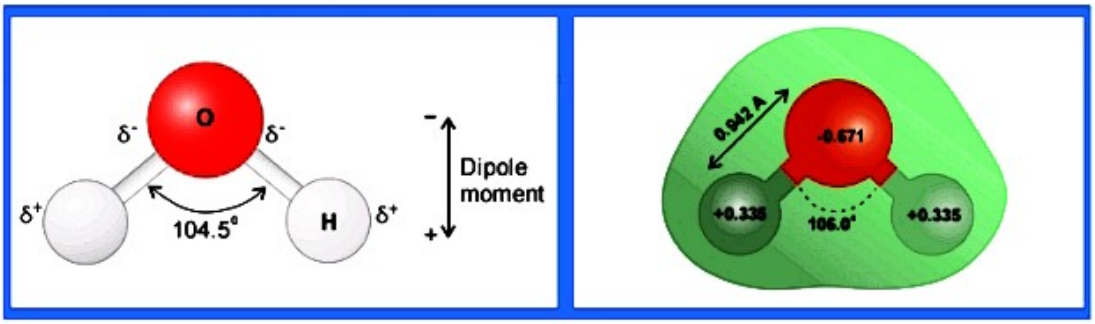
Water is a colourless and tasteless liquid. The unusual properties of water in the condensed phase (liquid and solid states) are due to the presence of extensive Hydrogen bonding between water molecules. This leads to high value of freezing point, boiling point, heat of vaporisation, surface tension, thermal conductivity, dipole moment and dielectric constant as comparison to other liquid.

Physical Properties of H_2O and D_2O

Heavy water (D_2O) has higher value of enthalpy of formation, enthalpy of vaporisation, density, melting point and boiling point in comparison to H_2O.

Structure of Water

In the gas phase water is a bent molecule with a bond angle of 104.5° and O–H bond length of 95.7 pm



(a) The bent structure of water(H_2O)

(b) The water molecule as a dipole & It is a **highly polar molecule**, (Its orbital overlap picture). In the liquid phase water molecules are connected together by Hydrogen bonds.

Structure of Ice

Ice has **a very well ordered three dimensional** Hydrogen **bonded structure**.

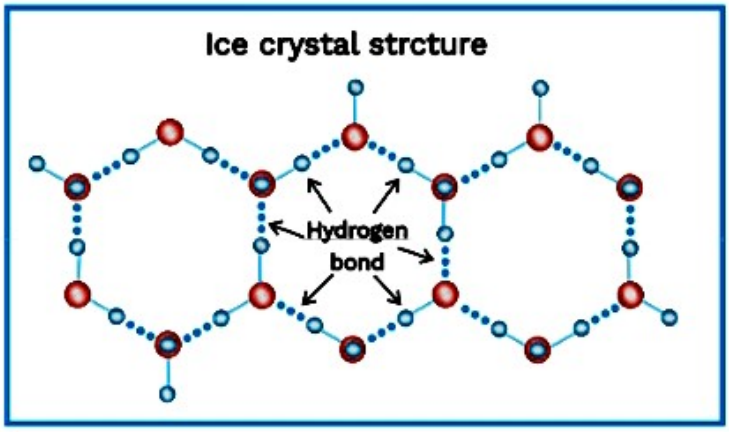
Observation

Examination of ice crystals with X-rays exhibit that each oxygen atom is surrounded tetrahedrally by four other hydrogen atoms at a distance of 276 pm. Hydrogen(H) bonding gives ice a rather open type structure with

Points to remember

Crystalline form of water is ice. At atmospheric pressure ice crystallises in the hexagonal form, however at extremely low temperature it condenses to cubic form. Density(d) of ice is less than that of water. Therefore, an ice cube floats of water(H_2O). In winter season ice produced on the surface of a lake provides thermal insulation which ensures the survival of the aquatic life.

wide holes. These holes can hold a few other molecules of appropriate size interstitially.



Q14 **Explain the amphoteric nature of water.**

Sol. Water can donate as well as accept a proton.

$$\underset{\text{Acid}}{H_2O} + \underset{\text{Base}}{H_2O} \longrightarrow H_3^+O + OH^-$$

Q15 **Why do lakes freeze top towards bottom ?**

Sol. The density of ice is less than of liquid water. So ice float on the surface. Thus the ice layer at lower temperature floats over the water below it. Due to this the freezing of water into ice takes place continuously from top towards bottom.

Q16 **Why is ice less dense than water and what kind of attractive forces must be overcome to melt ice?**

Sol. Ice has a cage like structure which has open spaces. In water each O-atom is bonded to four H-atoms (2 by H-bonds and 2 by covalent bonds) whereas in ice each O-atom is bonds to 3H-atoms. The open space in 3D structure of ice makes it less dense. To melt the ice H-bonding and lattice forces must be overcome.

Q17 **Can marine species live in distilled water ?**

Sol. No, distilled water does not have dissolved oxygen.

Q18 **Hydrogen oxide reacts with bases.**

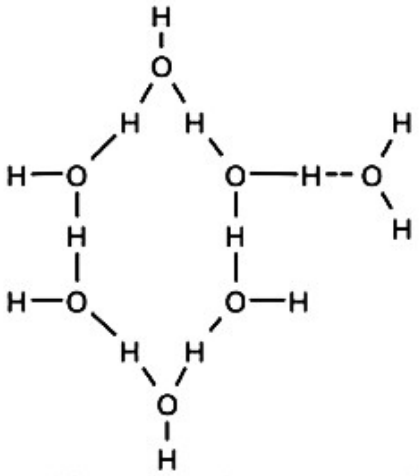
Sol. H_2O acts as as acid on treatment with base due to acidic nature of O – H bond.

Q19 **Ice floats over water. Why ?**

Sol. Ice possesses less density than water. In ice, molecules of H_2O are not packed closely but there exists vacant spaces in the crystal lattice and thereby leading to open cage like structure which ultimately leads to large volume and lower density.

Q20 **Explain, why the density of ice is less than water ? or why there is decrease in volume when ice melts ?**

Sol. In ice each molecule of H_2O is surrounded by three H_2O molecules in hexagonal honeycomb manner as shown in figure. On the other hand in water, each molecule is surrounded by four neighbouring molecules randomly. This arrangement gives rise to an open-cage like structure. There are evidently, a number of 'holes' or open spaces. In such a structure lesser number of molecules are packed per ml. When ice melts a large number of hydrogen bonds are broken. The molecules, therefore, move into the holes or open spaces and come closer to each other than they were in solid state. This results sharp increase in the density. Thus, ice has lower density than water and there is contraction in volume. The contraction continues form 0°C to 4°C as the hydrogen bonds are broken progressively. The density of water is maximum at 4°C.



Hexagonal honey comb structure of ice

(i) **Chemical properties:**

(i) with metal :

$$\underset{\text{Reactivity} > H_2}{\text{metal}} + H_2O \rightarrow H_2 \uparrow + \text{metal Hydroxide}$$

As Reactivity ↓ required temperature for Reaction ↑

$$Na + H_2O \rightarrow NaOH + H_2 \uparrow$$

$$Ca + H_2O \rightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2} + H_2 \uparrow$$

$$Mg + \text{Hot } H_2O \rightarrow \underset{\text{Magnesium hydroxide}}{Mg(OH)_2} + H_2 \uparrow$$

$$Zn + \underset{H_2O}{\text{Boiling water}} \rightarrow \underset{\text{Zinc hydroxide}}{Zn(OH)_2} + H_2 \uparrow$$

$$Fe + \text{Steam} \rightarrow Fe_2O_3 + H_2 \uparrow$$

$$[Al, Be, Pb] \xrightarrow{H_2O} \text{No reaction}$$

(Due to the formation of a protective layer of oxide)

(ii) **Reaction with non-metal**

No of covalent bond more than 1, so the more energy required to break the bond and non-polar also, and Interaction also week so, no Reaction Occur.

(iii) **Reactivity with Halogen**

$$F_2 > Cl_2 > Br_2 > \underset{\text{No Reaction}}{I_2}$$

$$F_2 + H_2O \rightarrow HF + O_2 \quad \text{(Displacement reaction)}$$

As F > O (Electronegativity)

If F_2 is in excess

$$O_2 + F_2 \longrightarrow OF_2$$

Ionic compounds + water

Oxide + H_2O → Hydroxide

$$O^{-2} + \overset{\oplus}{H}O\overset{\ominus}{H} \rightarrow OH^-$$

$$Na_2O + H_2O \rightarrow NaOH$$

$$Mg + H_2O \rightarrow Mg(OH)_2 \text{ milk of magnesia}$$

$$\underset{\text{quick line}}{CaO} \xrightarrow{H_2O} \underset{\text{Slaked line}}{Ca(OH)_2}(s) \xrightarrow{H_2O} \underset{\text{White Turbidity}}{Ca(OH)_2} \xrightarrow{H_2O} \underset{\text{Clear solution line water}}{Ca(OH)_2}$$

Nitride + H_2O

$N^{-3} + H^+ \ OH^- \rightarrow OH^- + NH_3$

$Ca_3N_2 + H_2O \rightarrow Ca(OH)_2 + NH_3 \uparrow$

Amide + H_2O

$NH_2 + H_2O \rightarrow OH^- + NH_3$

$NaNH_2 + H_2O \rightarrow NaOH + NH_3 \uparrow$

Imide (NH^{-2}) + H_2O

$Li_2NH + H_2O \rightarrow LiOH + NH_3 \uparrow$

Phosphide + H_2O

$P^{-3} + H^+ OH^- \rightarrow PH_3 + OH^-$

$Ca_3P_2 + H_2O \rightarrow Ca(OH)_2 + PH_3 \uparrow$ Phosphine

Peroxide + H_2O

$O_2^{2-} + H^+ OH^- \rightarrow H_2O_2 + OH^-$

$Na_2O_2 + H_2O \rightarrow NaOH + H_2O_2$

Hydride + H_2O

$H^+ + H_2O \rightarrow H_2 \uparrow + OH^-$

$CaH_2 + H_2O \rightarrow Ca(OH)_2 + H_2 \uparrow$

Superoxide + H_2O

$O_2^- + H_2O \rightarrow H_2O_2 + O_2 + OH^-$

O_2^- odd electron species unstable undergoes disproportionation reaction,

Example :

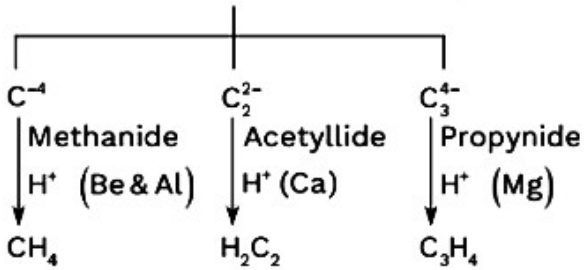
$KO_2 + H_2O \rightarrow KOH + H_2O_2 + O_2$

Points to remember

$\left[O_2 \xrightarrow{+e^-} O_2^- \xrightarrow{+e^-} O_2^{2-} \xrightarrow{+e^-} O^{-2}\right]$

CARBIDE + H_2O

Ionic carbide (s – block + Al, Zn)



$Ca^{2+}C_2^{-2} \rightarrow CaC_2 + H_2O \rightarrow Ca(OH)_2 + H_2C_2 \uparrow$

Q21 **Under what conditions of temp. and pressure, the formation of molecular Hydrogen(H) from atomic Hydrogen(H) will be favored most.**
(A) High temperature and high pressure.
(B) Low temperature and low pressure.
(C) High temperature and low pressure.
(D) Low temperature and high pressure.

Sol. **(D)**

$2H \longrightarrow H_2$ (Process exothermic)

The two Hydrogen(H) atoms convert into one H_2.

ΔS entropy $\downarrow$

$\Delta n_g \downarrow$

Then, increases the pressure the reaction moving from where pressure is low.

Q22 **In which property listed below Hydrogen(H) does not resemble alkali metals?**
(A) Tendency to form cation **(B) Nature of oxide**
(C) Combination with halogen **(D) Reducing character.**

Ans. **(B)**

Alkali metal $\longrightarrow$ Oxide (Basic in nature)

Hydrogen(H) $\longrightarrow$ Oxide (Amphoteric in nature)

Q23 **Hydrogen forms compounds with elements having atomic numbers: 9, 11, 12 and 17. What are their chemical formula? Compare their chemical behaviour.**

Sol.

$H_2 + F_2 \longrightarrow 2HF$ (Halogen acid, weak reducing agent)

$H_2 + 2Na \longrightarrow 2NaH$ (Ionic hydrides, reducing agent)

$H_2 + 2Mg \longrightarrow MgH_2$ (Polymeric hydrides, weak reducing agent)

$H_2 + Cl_2 \longrightarrow 2HCl$ (HCl stronger reducing agent than HF)

Q24 **Describe the industrial applications of hydrogen depends on:**
(a) The heat liberated when its atoms are made to combine on the surface of a metal.
(b) Its effect on unsaturated organic systems in presence of a catalyst.
(c) Its ability to combine with nitrogen under specific conditions.

Sol. **(a)** Atomic hydrogen welding torch.

(b) In preparation of saturated molecules;

$$CH_2 = CH_2 + H_2 \xrightarrow{\text{cat.}} CH_3CH_3$$

(c) Haber's process for NH_3.

Q25 **Hydrogen forms three types of bonds in the compounds. Describe each type of bonding using suitable examples.**

Sol. Ionic bond in NaH, covalent bond in H_2O, metallic bond in LaH_3.

Q26 **Ionic hydrides are frequently used to remove traces of water from organic compounds. What is the underlying basis of this process?**

Sol. H^- is a strong Bronsted base and thus it reacts with water easily.

Q27 **In the preparation of hydrogen in the laboratory from granular size, why the following acids cannot be used.**
(A) Conc. H_2SO_4 (B) Conc. HCl (C) Nitric acid

Sol. **(A)** Conc. H_2SO_4 is not used because a part of the acid is reduced to sulphur dioxide

$$\underset{(conc.)}{H_2SO_4} + H_2 \longrightarrow 2H_2O + SO_2$$

(B) Hydrogen liberated by the action of conc. HCl on size is impure as it contains fumes of volatile HCl. Also, $ZnCl_2$ formed, form a coating on zinc and the reaction stops after sometime.

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$$

(C) HNO_3 acts as an acid and an oxidising agent. The hydrogen (nascent) first formed reduces the nitric acid into various oxides.

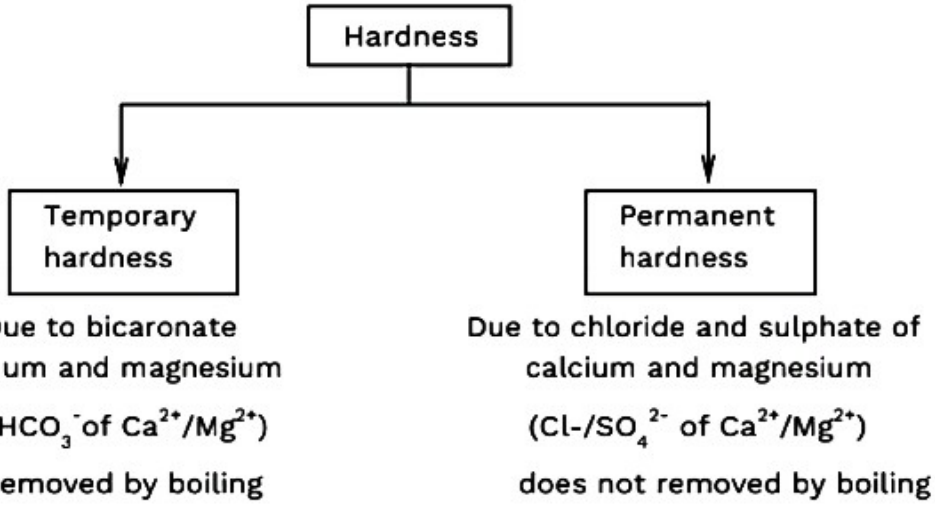
$$2HNO_3 + 2[H] \longrightarrow 2NO_2 + 2H_2O$$
$$2HNO_3 + 6[H] \longrightarrow 2NO + 4H_2O$$

Thus, the most suitable acid is dilute H_2SO_4.

"Hardness"

Rainwater is almost pure (may contain some dissolved gases from the atmosphere).it is a good solvent, at that time it flows on the surface of the earth, it dissolves several salts.

- The water that lathers with soap easily is called soft water.
  H_2O(soft water) $\xrightarrow{soap}$ lather
- The water that does not produce lather with soap is called hard water.
- Hardness in water is present due to soluble salt of (Ca^{2+} and Mg^{2+})
- Hard water does not produce lather because hard water contains magnesium and calcium salts.
  Soap $\xrightarrow{Ca^{2+}/Mg^{2+}}$ Insoluble compound. That's why lather does not produce by soap in hard water.



Removal of Temporary Hardness:

Temporary hardness is present due to presence of **magnesium and calcium hydrogen carbonates**.

(A) By Boiling

During boiling, the soluble $Mg(HCO_3)_2$ is transformed into insoluble $Mg(OH)_2$ and $Ca(HCO_3)_2$ is changed to insoluble $CaCO_3$. It is because of high-level solubility product of $Mg(OH)_2$ as compared to that of $MgCO_3$, that $Mg(OH)_2$ is precipitated. These precipitates can be eliminated by filtration.

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 \downarrow + CO_2 + H_2O$$

$$Mg(HCO_3)_2 \xrightarrow{\Delta} Mg(OH)_2 \downarrow + CO_2 + H_2O$$

Note:

Solubility in II A decreases generally except F^-, O^{-2}, OH^-, $C_2O_4^{-2}$.

Solubility → $\boxed{\begin{array}{l} CaCO_3 < MgCO_3 \\ Mg(OH)_2 < Ca(OH)_2 \end{array}}$

(B) Clark's Method :

In this method, calulated amount of lime is add up to hard water and this precipitates out calcium carbonate($CaCO_3$) and magnesium hydroxide which can be filtered off.

$$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$$

$$Mg(HCO_3)_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2 \downarrow + CaCO_3 \downarrow$$

All the reaction of temporary hardness is anion based and bicarbonate is the reason for Precipitate.

Q28 **Distinguish between :**
(i) Hard and soft water and
(ii) Temporary and permanent hardness.

Sol. **(i)** Hard water contains soluble salts of Ca^{2+} and Mg^{2+} so it does not give lather with soaps.

(ii) Temporary hardness is due to the presence of bicarbonate of Ca and Mg while permanent hardness is due to chlorides and sulphates of Ca and Mg. Temporary hardness can be removed simply by boiling but removal permanent hardness requires methods like use of Zeolites, Calgon.

Q29 **Temporary hard water become soft on boiling. Explain.**

Sol. Temporary hardness of water is due to the presence of soluble bicarbonate of calcium and magnesium. On boiling, the bicarbonates are converted into insoluble carbonates which can be removed by filtration.

$$Ca(HCO_3)_2 \longrightarrow CaCO_3 + H_2O + CO_2$$

$$\underset{\text{Soluble}}{Mg(HCO_3)_2} \longrightarrow \underset{\text{Insoluble}}{Mg(OH)_2} + H_2O + CO_2$$

Q30 **Soft water produces lather with soap but hard water, Why ?**

Sol. As hard water is associated with Ca^{2+} and Mg^{2+} ions but hard water forms lather only when all the Ca^{2+} and Mg^{2+} ions are precipitated out. On using soap in hard water, the ion $RCOO^-$ (present in soap RCOONa) responsible for lather formation is removed out in form of insoluble salts of Ca & Mg.

$$2RCOO^- + Ca^{2+} \longrightarrow (RCOO)_2\,Ca \text{ (insoluble)}$$

Removal of Permanent Hardness

(1) Na_2CO_3 (Washing soda):

Washing soda reacts with soluble calcium and magnesium chlorides and sulphates in hard water to form insoluble carbonates.

$$Ca^{2+} + Na_2CO_3 \longrightarrow CaCO_3\downarrow + Na^+$$
$$Mg^{2+} + Na_2CO_3 \longrightarrow MgCO_3\downarrow + Na^+$$

(2) Calogen's Process (Sodium Hexametaphosphate-$Na_6P_6O_{18}$):

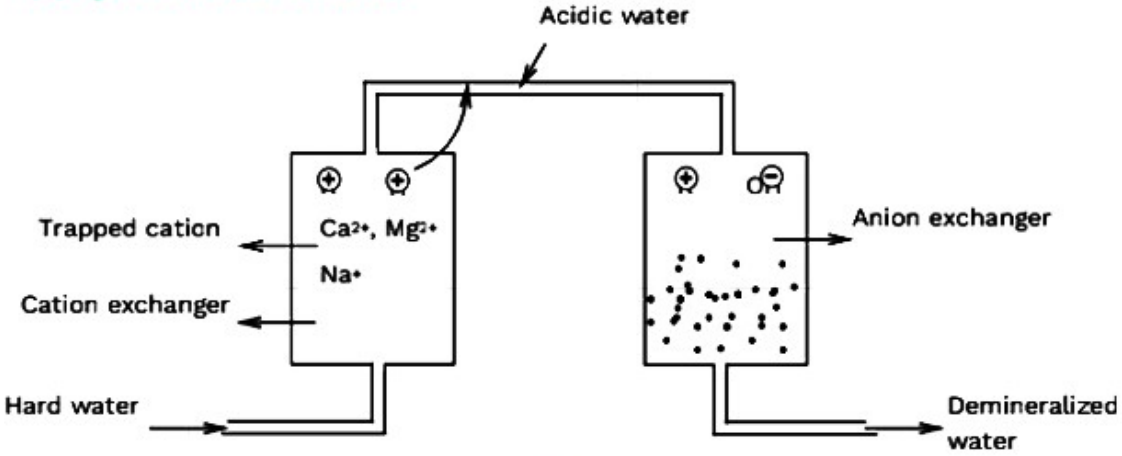
$$Na_6P_6O_{18} \longrightarrow 2Na^+ Na_4P_6O_{18}^{2-}$$
$$M^+ + \underset{\text{(Calogon)}}{Na_4P_6O_{18}^{-2}} \longrightarrow 2Na^+ \underset{\text{Part of anion}}{Na_2MP_6O_{18}^{2-}}$$

(3) Permutit / Zeolite / Ion exchange method

Sodium Zeolite is stone (Aluminium silicate)

$Na_2Z + M^{+2}\ (Ca^{2+}/Mg^{2+}) \longrightarrow MZ + 2Na^+$

(4) Synthetic Resin Method:



- Let us now explain the working of these resins in the demineralization of

hard water. First of all, hard water with cation exchange resins supported over granule. All the cations present in hard water will exchange with H^+ ions and R^- ions furnished by resins as shown Ca^{2+}, Mg^{2+} and Na^+.

- After the removal of cationic part this water becomes soften but it can not be used for drinking purpose because this water still contains the impurity of acid. To remove anionic part this acidic water is passed through anion exchanger.
- Then water is passed through the first tank to the second tank passed with acidic water and the second tank which is passed with anions exchange resins support over Granule.

With the help of this anion exchanger anions like Cl^-, SO_4^{-2} & HCO_3^- are removed.

- $R{-}NH_3^+OH^- + H^+ + Cl^- \rightarrow R{-}NH_3^+Cl^- + H^+OH^-$
- Now after removal of both cationic and anionic part this water is free from its hardness and impurities & it can be used for drinking purpose.

Q31 **A sample of hard water is permitted to go through the anion exchange resin.**
(A) Will it produce lather with soap easily ?
(B) Will it give test of chloride ion ?

Sol. **(A)** No ; Ca^{2+}, Mg^{2+} ions are still present.

(B) No; Cl^- ions are removed by anion exchange.

Q32 **Hard water is softened before use in boilers. Why ?**

Sol. Hard water contains salts of calcium and magnesium which gives precipitates of $CaCO_3$ and $MgCO_3$ along with $CaSO_4$ on heating which forms scales in boilers. So, it cannot be directly used in boilers. Hard water is softened before its use in boiler to avoid this scale formation.

H_2O_2 preparation:

Hydrogen peroxide is an important chemical used in pollution control treatment of domestic and industrial effluents.

(1) $BaO_2 + H_2SO_4 \longrightarrow H_2O_2 + BaSO_4\downarrow$

(2) $BaO_2 + H_3PO_4 \longrightarrow H_2O_2 + Ba_3(PO_4)_2\downarrow$

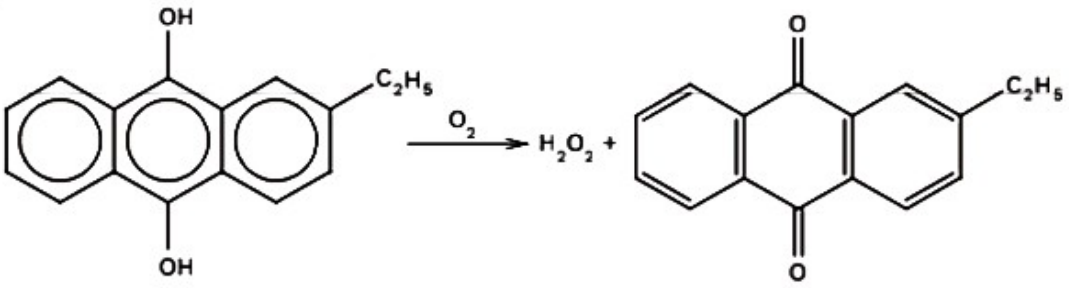
(3) $Na_2O_2 + ice \longrightarrow NaOH + H_2O_2$

(4) Electrolysis of SO_4^{2-}: Sulphate anion transfer to anode and oxidation in the anode sulphate are already in (+6) oxidation. There is not possible to oxidation of sulphate at a place of sulphate oxidation occur in Oxygen O^{-2}.

Anode: $2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e^-$

$S_2O_8^{2-} \xrightarrow{H_2O} SO_4^{2-} + H_2O_2$

(5)



2-ethylanthraquionol

Or $H_2 + O_2 \xrightarrow[\text{2-ethylanthraquinonl}]{\text{Catalyst}} H_2O_2$

H_2O_2 is unstable at room temperature. So, during purification/ concentration of H_2O_2 distillation under reduced pressure is used.

$$\underset{\text{dil.}}{H_2O_2(aq.)} \longrightarrow \underset{\text{Conc.}}{H_2O}\uparrow$$

So, on decreasing the pressure, H_2O gas reduced rapidly and this process is called distillation under reduced pressure.

Physical Properties:

	H_2O_2		H_2O
Strength of H-bond →	H_2O	<	H_2O
Melting point →	H_2O_2	<	H_2O
Boiling point →	H_2O_2	>	H_2O

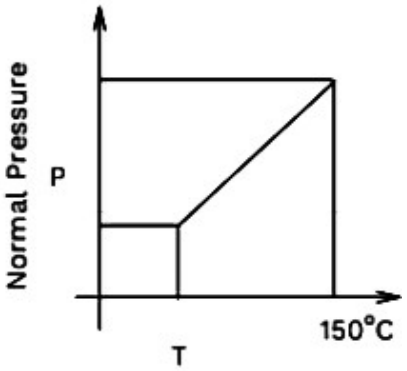
Pure H_2O_2 is a pale-blue liquid, acidic and oxidising as well as a reducing agent.

Extrapolation

Moving the graph forward and finding out the value is known as Extrapolation.

H_2O_2 boiling point → 150°

Boiling point of H_2O_2 is obtained by extrapolation.

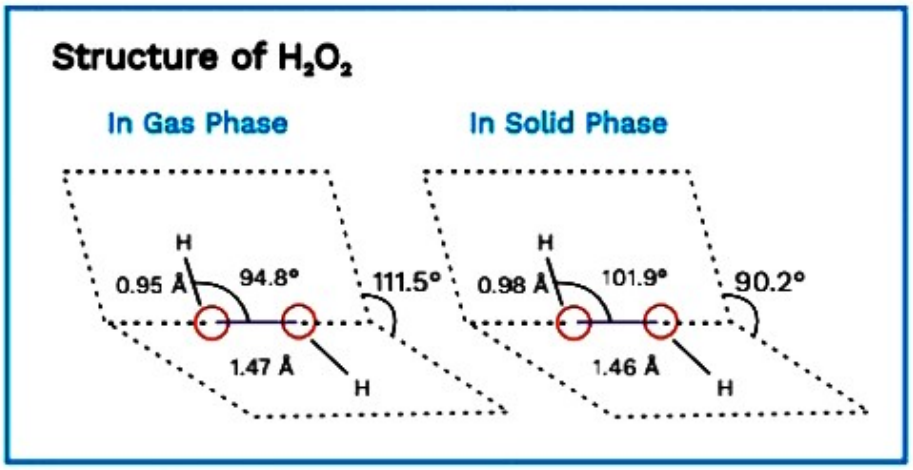


Structure:

Hydrogen peroxide has a non-planer open book like structure.

a) In gas phase structure its dihedral angle is 111.5°.

b) At 110K it is exist in solid phase and its dihedral angle become 90.2°.



Chemical Properties of H_2O_2:

It acts like both oxidising and reducing agent in acidic as well as in alkaline medium. Some useful reactions are given below.

(i) Oxidising action of H_2O_2 in acidic medium

$2Fe^{2+}(aq) + 2H^+(aq) + H_2O_2(aq) \longrightarrow 2Fe^{3+}(aq) + 2H_2O$

$PbS(s) + 4H_2O_2(aq) \longrightarrow PbSO_4(s) + 4H_2O$

(ii) Reducing action of H_2O_2 in acidic medium

$2MnO_4^- + 6H^+ + 5H_2O_2 \longrightarrow 2Mn^{2+} + 8H_2O + 5O_2$

$HOCl + H_2O_2 \longrightarrow H_3O^+ + Cl^- + O_2$

(iii) Oxidising action of H_2O_2 in basic medium

$$2Fe^{2+} + H_2O_2 \longrightarrow 2Fe^{3+} + 2OH^-$$
$$Mn^{2+} + H_2O_2 \longrightarrow Mn^{4+} + 2OH^-$$

(iv) Reducing action of H_2O_2 in basic medium

$$I_2 + H_2O_2 + 2OH^- \longrightarrow 2I^- + 2H_2O + O_2$$
$$2MnO_4^- + 3H_2O_2 \longrightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$$

Storage:

In the presence of light H_2O_2 decomposes slowly.

$$2H_2O_2 \longrightarrow 2H_2O + O_2(g)$$

- This reaction is catalysed by metal surfaces or traces of alkali (present in glass containers).
- As a stabiliser urea is used.
- Dust induces explosive decomposition of the compound so it is kept away from dust. Therefore, it is stored in plastic vessels or wax-lined glass in dark.

Uses

Now a days production of H_2O_2 has been increased drastically due to its wide scale use. Its uses are summarised in given table:

(i) It is used as a **hair bleach and also as a mild disinfectant in daily life**. In the market it is sold as **perhydrol in the form of an antiseptic.**

(ii) It is utilized in the industries as a **bleaching agent** for textiles, paper pulp, leather, oils, fats, etc.

(iii) It is used in the synthesis of hydroquinone, tartaric acid and certain food products and **pharmaceuticals (cephalosporin) etc.**

(iv) It is employed to manufacture chemicals like sodium perborate and per-carbonate, which are utilized in **high quality detergents.**

(v) It is also utilized in Environmental Chemistry. For example, in pollution control treatment of domestic and industrial effluents, oxidation of cyanides, restoration of aerobic conditions to sewage wastes, etc.

Q33 **Complete the following reactions :**

(A) $CaO_{(s)} + H_2O_{(l)} \longrightarrow$

(B) $Na_2O_{(s)} + H_2O_{(l)} \longrightarrow$

Sol. **(A)** $CaO_{(s)} + H_2O_{(l)} \longrightarrow Ca(OH)_{2(aq)}$

(B) $Na_2O_{(s)} + H_2O_{(l)} \longrightarrow 2Na(OH)_{(aq)}$

Q34 **Discuss the importance of heavy water in nuclear reactors.**

Sol. It serves as a moderator to slow down the fast-moving neutrons (essential for nuclear fission).